

Chinmay Sonar
PhD Student
[Website] [Scholar] [LinkdIn] [GitHub]

+1 (805) 895-4475
csonar@cs.ucsb.edu
6510 El Colegio Road, Goleta, CA 93117

PhD student working in algorithms with a strong mathematical foundation and an enthusiasm for programming

EDUCATION

- **University of California, Santa Barbara, USA** *Fall 2019 - Current*
PhD Student, Computer Science
Research: Design of exact and approximate algorithms for geometric problems
GPA: 4/4
Advisor: [Prof. Subhash Suri](#)
- **Indian Institute of Technology Gandhinagar, India** *Fall 2014-2019*
Master of Technology, Computer Science (2017-2019)
Research: Algorithms for problems in Computational Social Choice [\[Thesis\]](#)
GPA: 9.83/10
Advisor: [Prof. Neeldhara Misra](#)
Bachelor of Technology, Mechanical Engineering (2014 - 2017)

WORKING PAPERS

- **Anonymity Preserving Space Partitions**
with Úrsula Hébert-Johnson, Subhash Suri and Vaishali Surianarayanan
(manuscript submitted for publication)
- **Equitable Division of a Path**
with Neeldhara Misra, P. R. Vaidyanathan and Rohit Vaish
(manuscript submitted for publication) [\[ArXIV\]](#)

PUBLICATIONS

- **Fair Covering of Points by Balls**
with Daniel Lokshtanov, Subhash Suri and Jie Xue
In Canadian Conference of Computational Geometry (CCCG 2020), Saskatoon, Canada [\[pdf\]](#)
- **On the Complexity of Winner Verification and Candidate Winner for Multiwinner Voting Rules**
with Palash Dey and Neeldhara Misra
In International Joint Conference on Artificial Intelligence (IJCAI'20), Yokohama, Japan [\[pdf\]](#)
- **Robustness Radius for Chamberlin-Courant on Restricted Domains**
with Neeldhara Misra
In International Conference on Current Trends in Theory and Practice of Informatics (SOFSEM'19), Nový Smokovec, Slovakia [\[pdf\]](#)
- **On the Complexity of Chamberlin-Courant On Almost Structured Profiles**
with Neeldhara Misra and P. R. Vaidyanathan
In International Conference on Algorithmic Decision Theory (ADT'17), Luxembourg [\[pdf\]](#)
- **Design and development of an efficient Onion harvester for Indian farms**
with Amogh Parab, Prasannajeet Mane, Janga Sai Kiran, Panna Lal Saini, Vineet Vashista
In Machines, Mechanisms and Robotics 2019, Mumbai, India [\[pdf\]](#)

WORK EXPERIENCE

- **UCSB CS | Teaching Assistant:** (Oct 2019 - Present)
 - CS 130A and CS 130B: Data Structures and Algorithm Design and Analysis
 - Taught fundamental algorithmic paradigms in the lab sections with over 50 students
 - Assisted students for programming assignments in Python and C++
- **Undergraduate Research Intern, Clemson, SC, USA** (Summer 2017)
 - Developed a simulation software for resonance in hydrogel in C and C++
 - Created a user-friendly interface to study visualizations in Matlab and Mathematica
 - Simulation is used to study mechanical properties of gel by Kuksenok research [group](#) [\[poster\]](#)
- **Undergraduate Research Intern, IIT Gandhinagar, India** (Summer 2016)
 - Implemented popular multiwinner election rules in Python
 - Designed efficient algorithms for election winner determination resulting in publication at Algorithmic Decision Theory 2017 conference

RELEVANT GRADUATE COURSEWORK

- Graph Theoretic Algorithms
- Randomized Algorithms
- Combinatorics, Linear Algebra
- Computational Geometry
- Advanced Distributed Systems
- Blockchain and Cryptocurrencies
- Cryptography
- Machine Learning

TECH ARSENAL

- **Programming Languages.** Python (Proficient), C++ (Good), Shell Scripts (Good), Ruby (Good)
- **Operating Systems/ Tools.** Ubuntu, Windows, MATLAB, Mathematica, git workflow, $\text{\LaTeX}$

SELECTED PROJECTS AND ADDITIONAL EXPERIENCE

- **Anonymous Elections on Blockchain:** (Fall 2020)
 - Developed and implemented anonymous voting protocol to conceal the voter identities
 - Python implementation of the protocol scaled efficiently on a private ethereum network with only a linear time overhead compared to the conspicuous protocol [\[code\]](#)
- **Blockchain Model:** (Winter 2020)
 - Developed and implemented a distributed consensus protocol for distributed ledger (Blockchain)
 - 1000+ lines of code in Python with socket programming and multiprocessing [\[code\]](#)
- **Garbage Collection in Python:** (Spring 2020)

Studied and benchmarked garbage collectors in PyPy and CPython. Compared the performance of state of the art Garbage Collection algorithms. [\[code\]](#)
- **Electricity generation prediction model:** (in collaboration with govt. of India) (Spring 2019)

Built and compared various generation prediction model using time series analysis. With the train dataset of five years, model yielded an accuracy of over 75%.
- **IIT Gandhinagar CS | Teaching Assistant:**
 - ES 112 - Introduction to Python: Taught fundamentals of programming using python.
 - ES 214 - Discrete Mathematics: Taught fundamental mathematical concepts for CS.
 - ACM-W Summer School - Algorithmic Game Theory: Conducted problem solving sessions.

SCHOLARSHIPS AND AWARDS

- Graduate Entrance Fellowship (UCSB), 2019
- Selected among top 25 students in India for a pre-conference school fellowship at CALDAM 2019.
- Awarded *Swabhanu challenge travel grant to present outstanding undergraduate research* (IIT Gandhinagar) work in international conference (ADT) at Luxembourg, 2017
- Dean's List Award for excellent academic performance, IIT Gandhinagar, 2016 & 2017

MISCELLANEOUS

- **Professional Service:** Reviewer for SODA (2020), SoCG (2020), MFCS (2019), AAMAS (2019) conferences
- **CS Graduate Representative:** Co-lead CS talks and events [committee](#) (UCSB, 2019-20)
- **Competitive Programming:** Qualified for the *ACM-ICPC* India regionals as the University topper (2017) [\[YouTube\]](#)
- **Languages:** English (fluent), Hindi (fluent), Marathi (fluent), Spanish (beginner)